Be Your Own Datacenter — Challenge Write-Up

QVAC × PEARS (BY TETHER)

> Your modern digital life is increasingly centralized. Your data, compute, and daily tools are locked behind (and subject to) subscription clouds, external servers, and corporate intermediaries. At **QVAC** and **Pears, by Tether**, we believe true autonomy begins with your own hardware.
>
> **Your mission:** Build a solution that tackles an immediate problem for your university or local community using AI that runs 100% locally without inference cloud dependencies or external APIs. You can place the inference either at the device itself (on your machine) or on a device you control. Earn bonus points by cutting out middlemen entirely and distributing your solution through a secure, peer-to-peer architecture.

How the Challenge Works

Quick Flow

1. **Identify a High-Impact Problem:** Find an immediate problem affecting your campus or local community—such as offline campus assistance, collaborative study groups, privacy-first administrative records, or localized emergency tools.
2. **Mandatory Local Inference with QVAC:** All AI inference must run using **QVAC**. This can be accomplished directly via QVAC fabric running on your laptop/machine and exposing a local inference server, or running inference directly on-device. Zero external inference clouds or paid APIs (e.g., OpenAI, Anthropic) are permitted.
3. **Flexible Hardware Topology:** If your client device (e.g., a mobile phone) lacks the compute power for heavy inference, you can tunnel requests to a separate, more powerful machine that *you directly control* (such as your own laptop running QVAC fabric).
4. **Eliminate Intermediaries (Bonus Points):** Remove centralized relay servers or hosting providers. Gain significant bonus points by orchestrating direct peer-to-peer connections and tunneling using **Holesail (holesail.io)** and distributing application logic with **Pears**.
5. **Demo & Submission:** Submit your open repository, an architectural breakdown showing the data/inference path, setup steps, and a live working demo displaying local hardware execution.

Evaluation & Architectural Criteria

Component	Requirement	Description
QVAC Local Inference	Mandatory	Inference must run locally using QVAC (on-device or via QVAC fabric on a machine you own). No cloud APIs or SaaS endpoints allowed.
Holesail / P2P Tunneling	Bonus	Tunneling compute from secondary clients (e.g., mobile to laptop) peer-to-peer using **holesail.io** without centralized intermediaries.
Pears P2P Architecture	Bonus	Distributing data, states, or app updates directly peer-to-peer with Pears to maintain zero reliance on centralized infrastructure.
Community Utility & UX	Core Metric	Practical relevance of the problem addressed for the university or local community, speed of response, and ease of use.

Verification & Fair Play Rule:
All submitted projects must demonstrate end-to-end functionality in an isolated, offline environment or over an unproxied local/P2P network. Solutions observed routing inference payloads to closed cloud APIs will be disqualified from the track.